Saelyne Yang

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Research Interests: Human-AI Interaction; Human-Agent Collaboration; Multimodal Understanding; Procedural Task Learning; Skill Learning

Education

KAIST (Korea Advanced Institute of Science and Technology) — **Dajeon, S.Korea**
Ph.D. in School of Computing (Advisor: Juho Kim) — *Mar.2021 - Feb. 2026*

- Outstanding Thesis Award
- GPA: 4.16/4.3

KAIST (Korea Advanced Institute of Science and Technology) — **Dajeon, S.Korea**
M.S. in School of Computing (Advisor: Juho Kim) — *Mar.2019 - Feb.2021*

- Outstanding Thesis Award

KAIST (Korea Advanced Institute of Science and Technology) — **Dajeon, S.Korea**
B.S. in School of Computing — *Mar.2015 - Feb.2019*

- Graudated with Honors (Magna Cum Laude)
- Exchange student in UCSC, 2017 Summer

Employment

Carnegie Mellon University, Human Computer Interaction Institute — **Pittsburgh, PA**
Postdoctoral Fellow, Advisor: Jeffrey P. Bigham — *Mar. 2026 - Present*

Adobe Research - STORIE Lab — **Seattle, WA**
Research Intern, Mentors: Ding Li, Anh Truong — *May. 2024 - Aug. 2024*

Autodesk Research - HCI & Visualization Group — **Toronto, Ontario**
Research Intern, Mentors: Jo Vermeulen, Justin Matejka, George Fitzmaurice — *Jun. 2023 - Sep. 2023*

LG AI Research - Advanced ML Lab — **Seoul, S.Korea**
Research Intern, Mentor: Moontae Lee — *Oct. 2022 - Feb. 2023*

Adobe Research - Creative Technologies Lab — **Cambridge, MA (Virtual)**
Research Intern, Mentor: Hijung Valentin Shin — *Jun. 2020 - Sep. 2020*

ACRYL — **Seoul, S.Korea**
Software Engineer Intern — *Jan. 2018 - Feb. 2018*

Publications

Conference and Journal Papers

[c.14] GUIDE: A Benchmark for Understanding and Assisting Users in Open-Ended GUI Tasks
Saelyne Yang, *Jaesang Yu, Yi-Hao Peng, Kevin Qinghong Lin, Jae Won Cho, Yale Song, Juho Kim*
CVPR 2026

[c.13] "I Can't Keep Up": Accessibility Barriers in Video-Based Learning for Individuals with BIF
Hyehyun Chu, Seungju Kim, Chen Zhou, Yu-Kai Hung, ***Saelyne Yang*** *Hyun W. Ka, Juho Kim*
CHI 2026

[c.12] VideoMix: Aggregating How-To Videos for Task-Oriented Learning
Saelyne Yang, *Anh Truong, Juho Kim, Dingzeyu Li*
IUI 2025: ACM Conference on Intelligent User Interfaces

[c.11] Silent Impact: Tracking Tennis Shots from the Passive Arm
Junyong Park, ***Saelyne Yang***, *Sungho Jo*
UIST 2024: ACM Symposium on User Interface Software and Technology

[c.10] AQuA: Automated Question-Answering in Software Tutorial Videos with Visual Anchors
Saelyne Yang, *Jo Vermeulen, George Fitzmaurice, Justin Matejka*
CHI 2024: ACM Conference on Human Factors in Computing Systems

[c.9] ExpressEdit: Video Editing with Natural Language and Sketching
Bekzat Tilekbay, ***Saelyne Yang***, *Michal A. Lewkowicz, Alex Suryapranata, Juho Kim*
IUI 2024: ACM Conference on Intelligent User Interfaces

[c.8] YTCommentQA: Video Question Answerability in Instructional Videos
Saelyne Yang, *Sunghyun Park, Yunseok Jang, Moontae Lee*
AAAI 2024: Association for the Advancement of Artificial Intelligence

[c.7] Towards Content-based Pixel Retrieval in Revisited Oxford and Paris
Guoyuan An, Woo Jae Kim, ***Saelyne Yang****, Rong Li, Yuchi Huo, Sungeui Yoon*
ICCV 2023: International Conference on Computer Vision

[c.6] Beyond Instructions: A Taxonomy of Information Types in How-to Videos
Saelyne Yang, *Sangkyung Kwak*, Juhoon Lee*, Juho Kim*
CHI 2023: ACM Conference on Human Factors in Computing Systems (*=equal contribution)

[c.5] AVscript: Accessible Video Editing with Audio-Visual Scripts
Mina Huh, ***Saelyne Yang****, Yi-Hao Peng, Xiang 'Anthony' Chen, Young-Ho Kim, Amy Pavel*
CHI 2023: ACM Conference on Human Factors in Computing Systems

[c.4] CatchLive: Real-time Summarization of Live Streams with Stream Content and Interaction Data
Saelyne Yang, *Jisu Yim, Juho Kim, Hijung Valentina Shin*
CHI 2022: ACM Conference on Human Factors in Computing Systems

[c.3] SoftVideo: Improving the Learning Experience of Software Tutorial Videos with Collective Interaction Data
Saelyne Yang, *Jisu Yim, Aitolkyn Baigutanova, Seoyoung Kim, Minsuk Chang, Juho Kim*
IUI 2022: ACM Conference on Intelligent User Interfaces

[c.2] Snapstream: Snapshot-based Interaction in Live Streaming for Visual Art
Saelyne Yang, *Changyoon Lee, Hijung Valentina Shin, Juho Kim*
CHI 2020: ACM Conference on Human Factors in Computing Systems

[c.1] What Makes It Hard for Users to Follow Software Tutorial Videos?
Saelyne Yang, *Juho Kim*
HCI Korea 2020, The HCI Society of Korea

Extended Abstracts

[p.3] Enhancing How People Learn Procedural Tasks Through How-to Videos
Saelyne Yang
UIST 2024 Doctoral Symposium

[p.2] Fine-Grained Action Understanding with Tools in Instructional Videos
Saelyne Yang, *Jaesang Yu, Jae Won Cho, Juho Kim*
CVPR 2024 Workshop on Learning from Procedural Videos and Language

[p.1] Improving Video Interfaces by Presenting Informational Units of Videos
Saelyne Yang, *Sankyung Kwak, Tae Soo Kim, Juho Kim*

Honors & Awards

Special Recognition for Outstanding Reviews (x2)
CHI 2025 *2025*

AAAI-24 Student Scholar
AAAI 2024 *2024*

Google Student Travel Grant
KAIST-Google *2023*

NAVER Ph.D. Fellowship
NAVER *2022*

Special Recognition for Outstanding Reviews
CSCW 2023 *2022*

Special Recognition for Outstanding Reviews
CHI 2022 *2022*

An Outstanding TA Award
School of Computing, KAIST *Spring 2022*

Inseong Fellow ($1,000)
KAIST, Scholarship given to students for their service and impact on the school *2021*

An Outstanding TA Award
School of Computing, KAIST *Spring 2021*

An Outstanding MS Thesis Award
School of Computing, KAIST *2021*

Kwanjeong scholarship ($20,000)
Scholarship given to outstanding graduate students in the nation *2019-20*

Encouragement Award
SW Talent Festival, hosted by the Ministry of Science and ICT *2018*

Dean's list
School of Computing, KAIST *Spring 2018*

Excellence Award
Writing Contest, KAIST *2018*

Google Women Techmakers Scholarship
Google APAC *2018*

LINE Scholarship
Scholarship given to selected students in the department *2017*

National Science and Technology Scholarship
Scholarship covered for undergraduate studies *2017-2018*

Academic Services

Organizing Committee
- UIST 2023 Video Previews Chairs

Program Committee
- UIST 2026
- AAAI 2026
- IUI 2025
- CVPR 2024 LPVL Workshop
- CHI 2022, 2023 Late-Breaking Work

Session Chair
- UIST 2025
- CHI 2024

Reviewer (* = Outstanding Reviews)
- CVPR 2026
- AAAI 2026
- CHI 2026, 2025(**), 2024, 2023, 2022(*), 2021, 2020
- UIST 2025, 2024, 2022
- IMWUT 2025
- IUI 2025
- CSCW 2023(*), 2020
- DIS 2025, 2023
- IEEE Transactions on Learning Technologies

Student Volunteer
- AAAI 2024
- IUI 2022
- CHI 2019

Invited Talks

Enhancing How People Learn Procedural Tasks Through How-to Videos
Twelve Labs *Jul, 2024*

Improving Video Experiences by Leveraging User Interaction and Content Analysis
University of British Columbia (Host: Prof. Dongwook Yoon) *Oct, 2023*

Improving Video Experiences by Leveraging User Interaction and Content Analysis
DGP Lab, University of Toronto *Sep, 2023*

Improving Video Experiences by Leveraging User Interaction and Content Analysis
UNIST - CSE333 Intro to HCI (Host: Prof. Jaeyeon Lee) *May, 2023*

HCI@KAIST Workshop Panel
Meta discussion on CHI, KAIST *Jun, 2022*

Teaching

Teaching Assistant
- **Wearable User Interface**
  KAIST CS486 *Spring 2024*
- **Introduction to Human-Computer Interaction**
  KAIST CS374 *Spring 2021, 2020, 2019*
- **Data Structure**
  KAIST CS206 *Spring 2023*

- **Introduction to Programming**
  KAIST CS101 *Spring 2022, Fall 2021*

Service and Leadership

KAIST HCI Lunch Meeting
Organizer *2024*

KAIST-Google ExploreCSR
Organizer *2021*

KAIST Committee on Social Inclusion
Student representative *2021-2022*